UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 9 Physics Final Exam Practice

Term 1 2026–2027 • Set A • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. After formulating a testable hypothesis, which action most directly tests it? [4 marks]

A. Carry out a controlled investigation
B. Choose the most popular explanation
C. Ignore conflicting measurements
D. Treat the hypothesis as a law

Reference: pp. 4–6; p. 8, Check Your Progress 1–3 | **Answer A**

2. A car is represented by a single dot when studying its journey. Why is this model useful? [4 marks]

A. It removes the need for a reference frame
B. It simplifies the motion while retaining relevant position information
C. It includes every detail of the engine
D. It proves that a car has no size

Reference: pp. 6–7 | **Answer B**

3. Which statement correctly compares a scientific law and a scientific theory? [4 marks]

A. A law describes a relationship; a theory explains observations
B. A theory is only an unsupported guess
C. A theory becomes a law simply by getting older
D. A law explains everything and needs no evidence

Reference: p. 7 | **Answer A**

4. Which prefix means a factor of 10^{-3}? [4 marks]

A. mega
B. milli
C. kilo
D. micro

Reference: pp. 10–11 | **Answer B**

5. Round 4.75 m − 0.4168 m using the correct rule for subtraction. [4 marks]

A. 4.33 m
B. 4 m
C. 4.3 m
D. 4.3332 m

Reference: p. 12, Check Your Progress 8(b) | **Answer A**

6. A bent micrometer gives highly repeatable readings but disagrees with a trusted standard. How should the readings be described? [4 marks]

A. Both accurate and precise
B. Accurate but imprecise
C. Precise but inaccurate
D. Neither repeatable nor precise

Reference: p. 14; p. 16, Check Your Progress 12 | **Answer C**

7. For a spring calibration, y = (0.08 cm/g)x + 13.7 cm. What is the mass for a total length of 15 cm? [4 marks]

A. About 29 g
B. About 16 g
C. About 1.3 g
D. About 188 g

Reference: p. 22, Check Your Progress 21 | **Answer B**

8. The particle positions in the figure were recorded at equal time intervals. What is happening? [2 marks]



A. The object is speeding up to the right
B. The object has constant speed
C. The object is stationary
D. The object is slowing down to the right

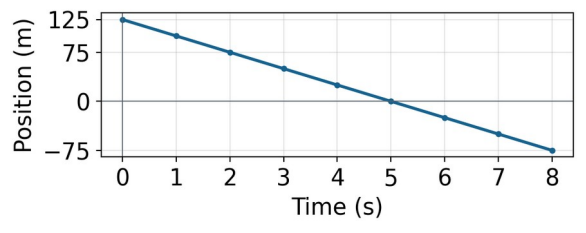
Reference: p. 31, Check Your Progress 3 | **Answer D**

9. Which pair contains only vector quantities? [4 marks]

A. Distance and speed
B. Mass and energy
C. Displacement and velocity
D. Work and pressure

Reference: p. 33 | **Answer C**

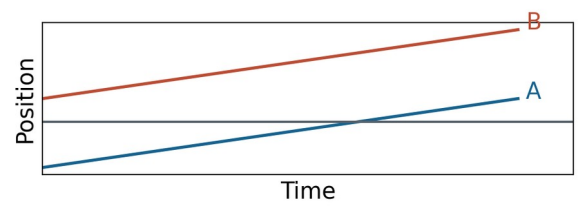
10. The car graph crosses x = 0 at 5.0 s. In which direction is the car moving? [4 marks]



A. The car is stationary
B. West with increasing speed
C. West at constant velocity
D. East at constant velocity

Reference: p. 39, Practice Problems 10–12 | **Answer C**

11. Pedestrians A and B have parallel position–time lines as shown. Which statement is correct? [4 marks]



A. Their velocities are equal
B. Their velocities have opposite signs
C. B must be moving faster because B is higher
D. A is stationary

Reference: p. 39, Practice Problem 13 | **Answer A**

12. Which statement describes instantaneous velocity? [4 marks]

A. A quantity with no direction
B. The same value as average velocity for every journey
C. Total distance divided by the full journey time
D. Velocity at a particular instant

Reference: pp. 43–44 | **Answer D**

13. An object has v = −6 m/s and a = −2 m/s². Its speed is: [4 marks]

A. Decreasing
B. Zero
C. Increasing
D. Unchanged

Reference: p. 56 | **Answer C**

14. A golf ball begins moving uphill at +2.0 m/s with a = −0.50 m/s². What is its velocity after 2.0 s? [4 marks]

A. −1.0 m/s
B. +1.0 m/s
C. +3.0 m/s
D. 0 m/s

Reference: p. 65, Practice Problem 16(a) | **Answer B**

15. An automobile accelerates from rest at 3.5 m/s². How far does it travel before reaching 25 m/s? [4 marks]

A. 7.1 m
B. 44 m
C. 179 m
D. 89 m

Reference: p. 68, Example Problem 4 | **Answer D**

16. At the highest point of a ball's vertical flight, neglecting air resistance, which is correct? [2 marks]

A. Velocity and acceleration are both zero
B. Speed equals 9.8 m/s
C. Acceleration is upward
D. Velocity is zero and acceleration is downward

Reference: pp. 71–74 | **Answer D**

Set A Section B Free Response

17. Three students measure the same spring. Their results are: student 1, (14.6 ± 0.2) cm; student 2, (14.8 ± 0.3) cm; student 3, (14.0 ± 0.1) cm. The accepted length is 14.8 cm. [10 marks]

Reference: pp. 11–14, Measurement and Figure 10

(a) Write the interval represented by each measurement. [3]

Answer Student 1: 14.4–14.8 cm [1]; student 2: 14.5 –15.1 cm [1]; student 3: 13.9–14.1 cm [1].

(b) Which student is most precise? Give a reason. [2]

Answer Student 3 [1]; smallest uncertainty, ±0.1 cm [1].

(c) Which student is most accurate? Explain. [2]

Answer Student 2 [1]; its central value equals the accepted 14.8 cm [1].

(d) Do students 1 and 2 agree within uncertainty? Does student 3 agree with them? [3]

Answer Students 1 and 2 agree [1]; their intervals overlap from 14.5 to 14.8 cm [1]. Student 3 has no overlap with either [1].

18. Take east as positive. A ship is initially 25 km east of point A and then travels east at 42 km/h for 1.5 h. Separately, a car changes velocity from 3.0 m/s west to 4.5 m/s east in 2.5 s. [10 marks]

Reference: p. 47, Practice Problem 32; p. 61, Practice Problem 8

(a) Calculate the ship's displacement. [2]

Answer $\Delta x = vt = 42 \times 1.5$ [1] = +63 km, east [1].

(b) Find the ship's final position relative to A. [2]

Answer $x_f = x_i + \Delta x = 25 + 63$ [1] = +88 km, east of A [1].

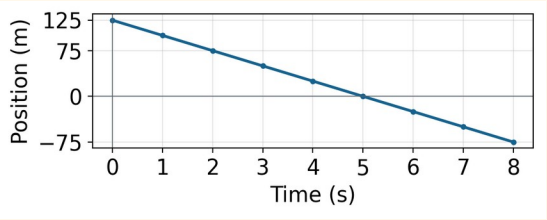
(c) State the difference between distance and displacement. [2]

Answer Distance is total path length, a scalar [1]. Displacement is the change in position, a vector [1].

(d) Calculate the car's average acceleration, including direction. [4]

Answer $v_i = -3.0$ m/s, $v_f = +4.5$ m/s [1]; $a = (v_f - v_i)/\Delta t$ [1] = 7.5/2.5 = 3.0 m/s² [1], east [1].

19. The graph shows a car moving along a straight line. Use the graph, including its axis units. [10 marks]



Reference: p. 39, Practice Problems 10–12, Figure 13

(a) Read the car's position at t = 0 s and t = 8 s. [2]

Answer x(0) = +125 m [1]; x(8) = −75 m [1].

(b) Calculate its displacement and total distance over 0–8 s. [3]

Answer $\Delta x = -75 - 125 = -200$ m [2: method and signed result]; distance = 200 m because there is no reversal [1].

(c) Determine its velocity from the graph. [3]

Answer v = slope = (−75 − 125)/(8 − 0) [1] = −25 m/s [1]; negative direction [1].

(d) At what time does it reach the origin? Does it stop there? [2]

Answer t = 5.0 s [1]. It does not stop: the slope stays negative and nonzero [1].

20. An elevator starts at rest, accelerates upward at 0.50 m/s² for 2.0 s, moves upward at 1.0 m/s for 12.0 s, then slows with downward acceleration 0.25 m/s² for 4.0 s. Take upward as positive. [10 marks]

Reference: p. 60, Practice Problem 4; pp. 57–59, 64–66

(a) Sketch a fully labelled velocity–time graph for the whole trip. [3]

Answer Axes t (s), v (m/s) [1]; straight rise (0,0) to (2,1), then horizontal to (14,1) [1]; straight fall to (18,0) [1].

(b) State the acceleration during each stage. [3]

Answer 0–2 s: +0.50 m/s² [1]; 2–14 s: 0 [1]; 14–18 s: −0.25 m/s² [1].

(c) Use graph areas to calculate the upward displacement. [3]

Answer ½(2)(1) + (12)(1) + ½(4)(1) [2 for three correct areas]; total 15 m upward [1].

(d) What physical quantity does the signed area under a velocity–time graph give? [1]

Answer Displacement [1].

